
Time-varying Neural Granger Causality

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Motivation

Many important applications are naturally represented as multivariate time series, including financial data such as stocks and physiological data such as EEG. Time-series modeling is central across such domains, and deep learning for time series has become a major area of study [1]. In these applications, it is not enough to model each variable in isolation; one wants to understand the dependence structure between variables.

Correlation is often too weak an object for this purpose, because it is symmetric and does not describe directed predictive influence. A more informative notion is Granger causality: one variable Granger-causes another if its past improves prediction of the other beyond what is achievable from the other variables' past alone. Neural Granger Causality explicitly builds nonlinear models for this purpose, showing that directed predictive dependence can be extracted from structured neural networks rather than only from linear VAR models [2]. Recent work also studies statistically principled neural tests for Granger causality, reinforcing that neural methods are not merely predictive black boxes but can support interpretable Granger-style inference [3].

However, nearly all of this machinery is most natural when the underlying directed dependency structure is time-invariant. In realistic data, that assumption is often too strong. In EEG, task changes or cognitive-state shifts can alter effective dependencies between channels. In financial data, market conditions, trends, and volatility regimes can alter predictive relations between assets. A natural relaxation is to assume not arbitrary continuous change, but piecewise time-invariant structure: within each regime, a fixed Granger-causal graph is meaningful, while between regimes the graph may change.

This immediately raises the segmentation problem. If the regime boundary were known, one could simply apply existing time-invariant methods locally. If regime lengths were fixed, the problem would largely reduce to identifying that length. The more interesting and realistic case is when boundaries are irregular and unknown. Since that fully general setting is difficult, we begin with the analytically clean base case of two regimes and one unknown change point. Solving this base case would provide both a practical segmentation tool and a foundation for subsequent extensions to multiple unknown regimes.

Approach Proposal

We study the following restricted but meaningful version of time-varying Granger-causal discovery. Assume a multivariate time series is generated by two regimes, each with a time-invariant directed dependency structure, and that the boundary between the two regimes is unknown. The objective is to estimate the most likely change point, then apply existing time-invariant Granger-causal discovery methods separately on the two inferred segments.

The logic is as follows. Time-invariant Granger-causal discovery is already well developed: linear methods such as VAR recover directed predictive dependence under linear assumptions, while neural methods extend this to nonlinear settings. In particular, neural-network-based Granger discovery

can be made interpretable through structured sparsity, as in Neural Granger Causality, and recent testing-based work further develops statistically grounded neural approaches for Granger inference [2, 3]. Thus, if the regime boundary were known, the problem would reduce to applying these local methods on each segment.

The difficulty is that the boundary is not known. A fully general setting with many irregular and unknown regimes is too broad for an initial study, so we begin with the simplest nontrivial case: two regimes and one unknown change point.

We propose a two-step approach. The first step is to learn a boundary score over candidate times t , interpreted as how likely t is to be the true regime boundary. Operationally, we fit a forward autoregressive model and a backward one and compute a score for t :

$$s(t) = \left| \frac{1}{h} \sum_{u=t-h+1}^t \|x_u - \hat{x}_u^{\text{fwd}}\|^2 - \frac{1}{h} \sum_{u=t+1}^{t+h} \|x_u - \hat{x}_u^{\text{bwd}}\|^2 \right| \quad (1)$$

The predicted boundary is then

$$\hat{\tau} = \arg \max_t s(t) \quad (2)$$

After estimating $\hat{\tau}$, the second step is to apply time-invariant Granger-causal discovery methods separately to $x_{1:\hat{\tau}}$ and $x_{\hat{\tau}+1:T}$ using highest-performing neural sequence models.

This formulation is intended as the first step toward the more general problem with more than two regimes. If later the number of regimes is assumed known, one natural extension is to detect multiple boundaries by selecting the top $k - 1$ sufficiently separated peaks of the boundary score. If the number of regimes is unknown, then additional model selection or regularization will be required, which we leave for subsequent analysis.

Data Synthesis

We generate multivariate time series from two piecewise time-invariant regimes. A change point τ is sampled uniformly from $\{L_{\min}, \dots, T - L_{\min}\}$. The first segment $x_{1:\tau}$ is generated from one dynamical system, and the second segment $x_{\tau+1:T}$ from a different system. In particular, we simulate nonlinear dynamics using Lorenz-96 with two distinct parameter settings, ensuring that the underlying dependency structure differs across regimes while keeping marginal statistics comparable. Optional Gaussian noise is added to control difficulty.

We additionally use TimeGraph as a benchmark dataset for time-series causal discovery, allowing us to evaluate whether the proposed boundary-detection approach generalizes beyond controlled simulations to standardized synthetic settings.

Relevant Papers and Baselines

Baselines:

- **Linear Time-invariant Baseline:** Classical VAR-based Granger causality
- **Neural Time-invariant Baseline:** Autoregressive MLP-based Granger causality
- Window-based distributional boundary baseline
- Midpoint or random boundary baseline

Papers:

- Tank et al. (2021): Neural Granger Causality [2]. This paper provides the foundation of using autoregressive cMLP and cLSTM to detect nonlinear, time-invariant Granger causality.
- Ma et al. (2024): Survey on time-series pre-trained models [1]. This paper offers already existing models, and alternative neural network architectures to consider.
- Hui et al. (2025): Doubly Robust Neural Test for Granger causality [3]. This paper provides statistical grounding for a full NN-based hypothesis testing procedure for Granger causality.

Plan

Expected (achievable):

- Generate synthetic multivariate time series with two regimes and a single change point (e.g., piecewise Lorenz-96).
- Implement forward and backward autoregressive predictors and compute the boundary score based on residual asymmetry.
- Evaluate boundary localization accuracy under varying conditions.
- Compare against a baseline change-point detection method.
- After estimating the boundary, apply RNN-based time-invariant Granger causal discovery methods separately on each segment and evaluate recovery accuracy.
- Compare against VAR and MLP baselines.

Moon-shot:

- To improve robustness of score computation, the boundary score can be smoothed across neighboring time points.
- To improve robustness of neural Granger causality detection, use statistically grounded post-training weight pruning rather than heuristics.
- Extend the boundary detection method to multiple unknown regimes by identifying multiple peaks in the boundary score.
- Develop a unified framework that jointly models boundary detection and dependency structure learning.
- Evaluate against the TimeGraph benchmark as well.

Evaluation Protocols and Metrics

- **Boundary localization error:** normalized absolute deviation with tolerance level $\delta > 0$:

$$\frac{|\hat{\tau} - \tau|}{T} \mathbb{I} \{ |\hat{\tau} - \tau| \geq \delta \}$$

- **Normalized Structural Hamming Distance (nSHD):** number of edge additions, deletions, and reversals between estimated Granger causality matrix and true Granger causality matrix:

$$\text{nSHD}(A, \hat{A}) = \frac{1}{d(d-1)} (\# \text{additions} + \# \text{deletions} + \# \text{reversals})$$

Even better, a reversal should have a penalty of at most 1, not 2.

- **Edge precision / recall / F1:** accuracy of recovered directed dependencies:

$$\text{TP} = \langle A, \hat{A} \rangle_F, \quad \text{PP} = \langle \hat{A}, \mathbf{1} \rangle_F, \quad \text{AP} = \langle A, \mathbf{1} \rangle_F \quad (3)$$

$$\text{Precision} = \frac{\langle A, \hat{A} \rangle_F}{\langle \hat{A}, \mathbf{1} \rangle_F}, \quad \text{Recall} = \frac{\langle A, \hat{A} \rangle_F}{\langle A, \mathbf{1} \rangle_F} \quad (4)$$

$$\text{F1} = \frac{2 \langle A, \hat{A} \rangle_F}{\langle A, \mathbf{1} \rangle_F + \langle \hat{A}, \mathbf{1} \rangle_F} \quad (5)$$

$$\text{Accuracy} = \frac{\langle A, \hat{A} \rangle_F + \langle \mathbf{1} - A, \mathbf{1} - \hat{A} \rangle_F}{\langle \mathbf{1}, \mathbf{1} \rangle_F} \quad (6)$$

References

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